24 + 1 = 25

Excellent

Blessed G/Michael Catholic Secondary School

First Semester Physics Test One for Grade 11

Oct. 2018 E.C Time Allowed: 1:25hr

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I. *Write "True" if the satmnet is correct or write "False" if the statement is wrong.[1 point each]*

False 1. In the scientific method used in physics, a hypothesis must be falsifiable through experimental testing or observation.

False 2. Knowledge acquire through experience: seeing something, hearing about something, feeling something called non-experential knowledge.

True 3. A unit vector is any vector that has a magnitude of exactly one, regardless of its direction.

True 4. Dot Product of a vector with itself is equal to the square of its length.

false 5. Cross Product of a vector with itself is equal to the square of the same vector.

II. *Choose the best answer from the given alternative.[1 point each]*

A 6. Which of the following is like a sixth sense that looks into the most intimate parts of our minds, which allows us to inspect how we are feeling and how our thoughts are operating.

A. Introspection B. Memory C. Sensory perception D. Testimony

D 7. Why is physics important to society?

A. It helps us understand the natural world
B. It improves technology and innovations
C. It contributes to advancements in medicine
D. All of the above

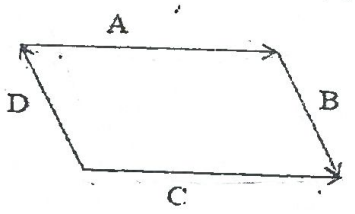
B 8. Identify the following quantities as scalar or vectors: the speed of a snail, the time it takes to run a killometer, and the free-fall acceleration.

Scalar Scalar Vector

A. Vector, scalar, scalar
B. Scalar, scalar, vector
C. Vector, scalar, vector
D. Scalar, vector, vector

B 9. Which one of the following is correct about the following parallelogram combination of vectors ?

A. $A + B = C + D$
B. $A - B = C + D$
C. $A + B - C = D$
D. $A - C + B = D$



A 10. Given three vectors $\vec{A}$, $\vec{B}$ *and* $\vec{C}$ such that $\vec{A} = (i + 2j)$m, $\vec{B} = (4i - j)$m and $\vec{C}$ unknown vector, determine the magnitude and direction of $\vec{C}$ for $-\vec{A} + \vec{B} - \vec{C} = 0$?

A. $\vec{C} = (3i + 3j)$m
B. $\vec{C} = (-3i + 3j)$m
C. $\vec{C} = (5i + j)$m
D. $\vec{C} = (-5i + j)$m

D 11. If two vectors $\vec{A} = 2i + 4j - 2k$ and $\vec{B} = 3i - 2j + nk$ are at right angles to each other, then the value of "n" must be:

A. 2 B. -2 C. 1 D. -1

C 12. The unit vector along $i + 2j + k$ *is:*

A. k B. $i + j$ C. $\frac{i+j}{\sqrt{2}}$ D. $\frac{i+j}{2}$

C 13. Find the unit vectors that are perpendicular to both $i + 2j + k$ and $3i - 4j + 2k$.

A. $\frac{1}{\sqrt{165}}(8i - j - 10k)$

B. $\frac{1}{\sqrt{6}}(i + 2j + 2k)$

C. $\frac{1}{\sqrt{29}}(3i - 4j + 2k)$

D. $\frac{1}{\sqrt{29}}(4i - 2j + 3k)$

D 14. If two vectors $\vec{A} = ai + 4j - 2k$ and $\vec{B} = 3i - bj + 2k$ are at parallel to each other, then the value of a & b must be: respectively

A. 3, 4 B. -4, -3 C. -4, 3 D. -3, 4

A 15. If $a \times i = 0$ *and* $a \cdot j = 0$; then a

A. Is perpendicular to $i \times j$

B. Is parallel to $i \times j$

C. Is equal to $\hat{k}$

D. Equal to zero

C 16. Which one of the following is correct?

A. If a, b, and c are three non-zero vectors, such that $a \cdot b = a \cdot c$ then it is true that b = c.

B. If a, b, and c are three non-zero vectors, such that $a \times b = c \times a$ then it is true that b = c

C. For any two vectors and U *and* V, $|U \times V| = |V \times U|$

D. If the direction of the vector is same as the direction of the vector , then $U \cdot V = |U|\,|V|$

D 17. Three forces act on a point: 3 N at 0°, 4 N at 90°, and 5 N at 217°. What is the net force?

$\sin 37^0 = \cos 53^0 = 0.6$ *and* $\sin 53^0 = \cos 37^0 = 0.8$

A. 0 N B. 1 N at 45^0 C. $\sqrt{2}$ N at 45^0 D. $\sqrt{2}$ N at 135^0

A 18. A vector $\vec{B}$, when added to the vector $\vec{C} = 3.0i + 4.0j$, yields a resultant vector that is in the positive y direction and has a magnitude equal to that of $\vec{C}$. What is the magnitude of $\vec{B}$?

A. 3.2 unit B. 3.5 unit C. 5.0 unit D. 9.0 unit

B 19. Vector A and B are non-zero vectors. The magnitude of A+B is four time the magnitude of A-B. then what is the angle between vector A and B?

A. $\cos^{-1}(5/12)$ B. $\cos^{-1}(3/4)$ C. $\cos^{-1}(4/3)$ D. $\cos^{-1}(12/5)$

III. *Short answer (Write only final answer).*

❖ ***Use for question number 21-24***

$A = i + j + k$

$B = 2i + 2j - 3k$

1. $\vec{B} + 3\vec{A} =$ $5i + 5j$
2. $\vec{A} \cdot 2\vec{B} =$ 2
3. $\vec{B} \times \vec{A} =$ $5i - 5j$
4. The unit vector along vector $\vec{A}$ is: $\frac{1}{\sqrt{3}}(i + j + k)$

IV. *Work out [show all necessary steps].*

5. Find angle between vector $A = (i + j)$ and vector $B = (2i - 2j)$?

$\cos\theta = \frac{A \cdot B}{|A||B|} = \frac{(1)(2) + (1)(-2)}{\sqrt{2}\sqrt{8}}$

② $A \cdot B = A_x B_x + A_y B_y = (1)(2) + (1)(-2)$

$= 2 - 2 = 0$

$|A| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$|B| = \sqrt{4 + 4} = \sqrt{8}$

$\cos\theta = 0 \Rightarrow \theta = \cos^{-1}(0)$

$= 90^\circ$

$\therefore \theta = 90^\circ$